TL01 PCA

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1 Introducción

Objetivo: aplicar PCA para reducir la dimensión de los datos; imágenes principalmente

PCA con scikit-learn:

- El método directo (y eficiente) para calcular PCA se basa en hallar la SVD "completa" de la matriz de datos
- La implementación de PCA en scikit-learn emplea un método aleatorizado, más eficiente que el directo

Aplicación a imágenes:

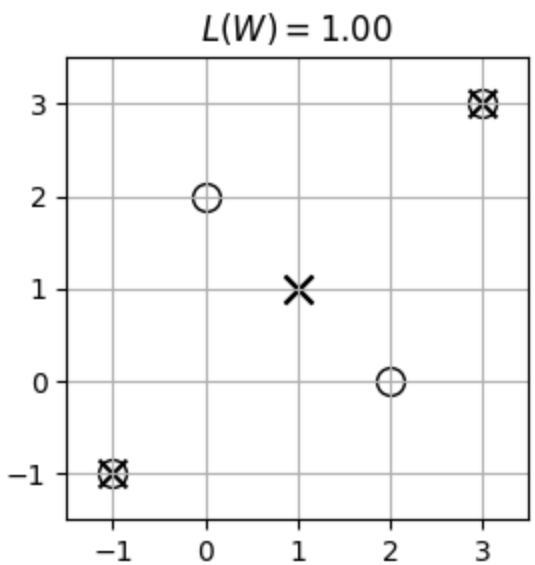
- Usaremos datasets muy populares: MNIST, Fashion-MNIST y CIFAR-10
- La dimensión de los datos en el caso de imágenes suele ser muy grande, lo que dificulta el aprendizaje de clasificadores
- Una aproximación convencional para facilitar el aprendizaje consiste en reducir previamente la dimensión de los datos
- Estudiaremos el error de reconstrucción en función de la dimensión reducida K
- Interpretaremos las componentes principales como imágenes base (eigenimages)
- Reconstruiremos imágenes y las compararemos con sus originales
- Estudiaremos la precisión de regresión logística en función de K
- Aunque trabajemos con imágenes, se puede proceder de forma parecida con otro tipo de datos (de alta dimensión)

2 PCA con scikit-learn

Método directo para aplicar PCA: a partir de la SVD "completa" de la matriz de datos

Ejemplo: $N = 4$ datos de $D = 2$ dimensiones que queremos reducir a $K = 1$ dimensión

In [ ]:
```
import numpy as np; import matplotlib.pyplot as plt
X = np.array([ [-1, -1], [0, 2], [2, 0], [3, 3] ]); N = len(X);
K = 1; _, _, Vt = np.linalg.svd(X - X.mean(0)); W = Vt[:K, :].T; Z = X @ W; hX = Z @ W.T
L = np.square(X - hX).sum(axis=1).mean()
fig, ax = plt.subplots(figsize=(3, 3)); ax.set_aspect("equal")
plt.axis([-1.5, 3.5, -1.5, 3.5]); plt.grid(True); ax.set_title(f'$L(W)={L:.2f}$')
plt.scatter(*X.T, facecolor='white', edgecolor='k', s=100)
plt.scatter(*hX.T, facecolor='black', s=100, marker='x');
```

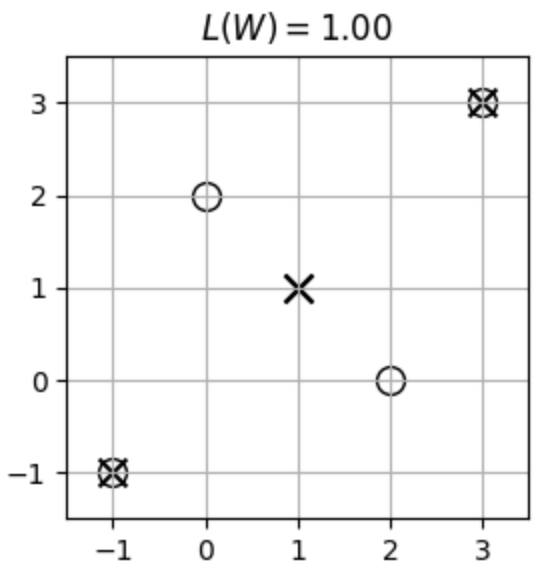


PCA con scikit-learn: emplea un método aleatorizado para matrices grandes, más eficiente que el directo

- **API:** https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html
- **Parámetros:** `n_components`, `svd_solver`, `random_state`, etc.
- **Atributos:** `components_`, `singular_values_`, `mean_`, `n_components`, etc.
- **Métodos:** `fit(X)`, `transform(X)`, `fit_transform(X)`, `inverse_transform(X)`, etc.

Ejemplo (cont.): con scikit-learn, `n_components` $= K$

In [ ]:
```
import numpy as np; import matplotlib.pyplot as plt; from sklearn.decomposition import PCA
X = np.array([ [-1, -1], [0, 2], [2, 0], [3, 3] ]); N = len(X)
K = 1; pca = PCA(n_components=K).fit(X - X.mean(0)); Z = pca.transform(X)
hX = pca.inverse_transform(Z); L = np.square(X - hX).sum(axis=1).mean()
fig, ax = plt.subplots(figsize=(3, 3)); ax.set_aspect("equal")
plt.axis([-1.5, 3.5, -1.5, 3.5]); plt.grid(True); ax.set_title(f'$L(W)={L:.2f}$')
plt.scatter(*X.T, facecolor='white', edgecolor='k', s=100)
plt.scatter(*hX.T, facecolor='black', s=100, marker='x');
```



3 MNIST

Modified NIST (MNIST): dataset de $70\,000$ imágenes 28×28 en gris de dígitos manuscritos

Muy popular: desde su introducción en los 90, MNIST ha sido muy usado para comparar técnicas de ML

Partición estándar: $60\,000$ primeras muestras para training y $10\,000$ restantes para test

In [ ]:
```python
import numpy as np; import datasets
ds = datasets.load_dataset("ylecun/mnist").with_format("numpy"); ds
```

Out[ ]:
```
DatasetDict({
    train: Dataset({
        features: ['image', 'label'],
        num_rows: 60000
    })
    test: Dataset({
        features: ['image', 'label'],
        num_rows: 10000
    })
})
```

In [ ]:
```python
train0 = ds['train'][0]
train0['image'].shape, train0['image'].dtype, train0['label']
```

Out[ ]:
```
((28, 28), dtype('uint8'), np.int64(5))
```

Agotado: pues ya se han alcanzado tasas de error muy reducidas, por debajo del 0.1%

In [ ]:

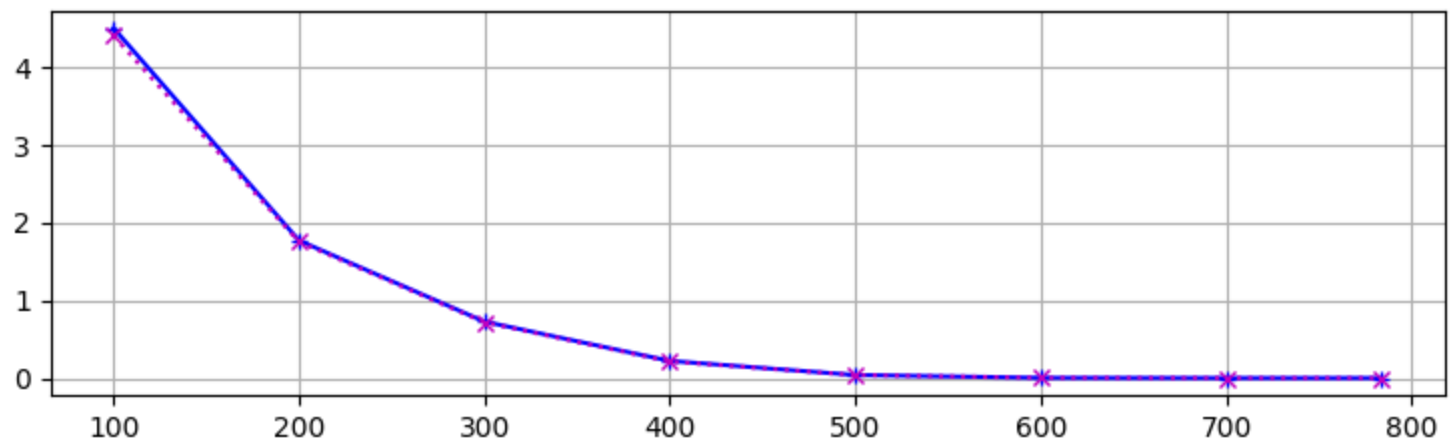
```
import matplotlib.pyplot as plt; nrows = 10; ncols = 15; N = nrows * ncols
_, axs = plt.subplots(nrows=nrows, ncols=ncols, figsize=(16, 16*nrows/ncols), constrained_layout=True)
for ax, x, y in zip(axs.flat, ds['train'][:N]['image'], ds['train'][:N]['label']):
    ax.set_axis_off(); ax.imshow(x, cmap=plt.cm.gray_r, interpolation="none"); ax.set_title(y)
```

4 PCA aplicado a MNIST

Error de reconstrucción: estudio en función de K

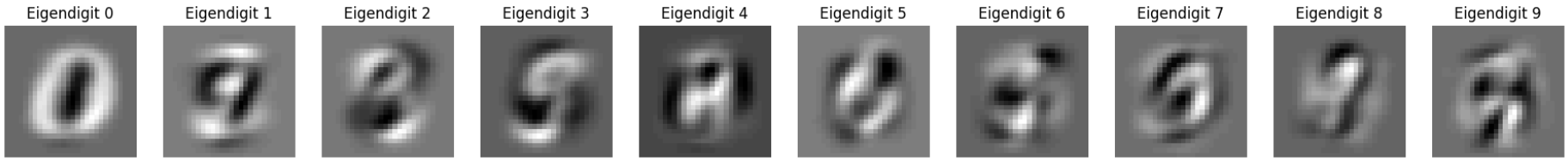
In [ ]:
```
import numpy as np; import datasets
ds = datasets.load_dataset("ylecun/mnist").with_format("numpy")
X_train = ds['train'][:]['image'].astype(np.float32).reshape(-1, 28*28) / 255.
y_train = ds['train'][:]['label'].astype(np.uint8)
X_test = ds['test'][:]['image'].astype(np.float32).reshape(-1, 28*28) / 255.
y_test = ds['test'][:]['label'].astype(np.uint8)
```

In [ ]:
```
import matplotlib.pyplot as plt; from sklearn.decomposition import PCA
max_K = np.min(X_train.shape); pca = PCA(n_components=max_K).fit(X_train)
Z_train = pca.transform(X_train); Z_test = pca.transform(X_test)
Ks = np.array([100, 200, 300, 400, 500, 600, 700, max_K])
L_train = np.empty_like(Ks, dtype=float); L_test = np.empty_like(Ks, dtype=float)
for i, K in enumerate(Ks):
    Z_train_K = Z_train.copy(); Z_train_K[:, K:] = 0.0; hX_train = pca.inverse_transform(Z_train_K)
    L_train[i] = np.square(X_train - hX_train).sum(axis=1).mean()
    Z_test_K = Z_test.copy(); Z_test_K[:, K:] = 0.0; hX_test = pca.inverse_transform(Z_test_K)
    L_test[i] = np.square(X_test - hX_test).sum(axis=1).mean()
plt.figure(figsize=(9, 2.5)); plt.grid(True); plt.plot(Ks, L_train, '+-b', Ks, L_test, 'x:m');
```

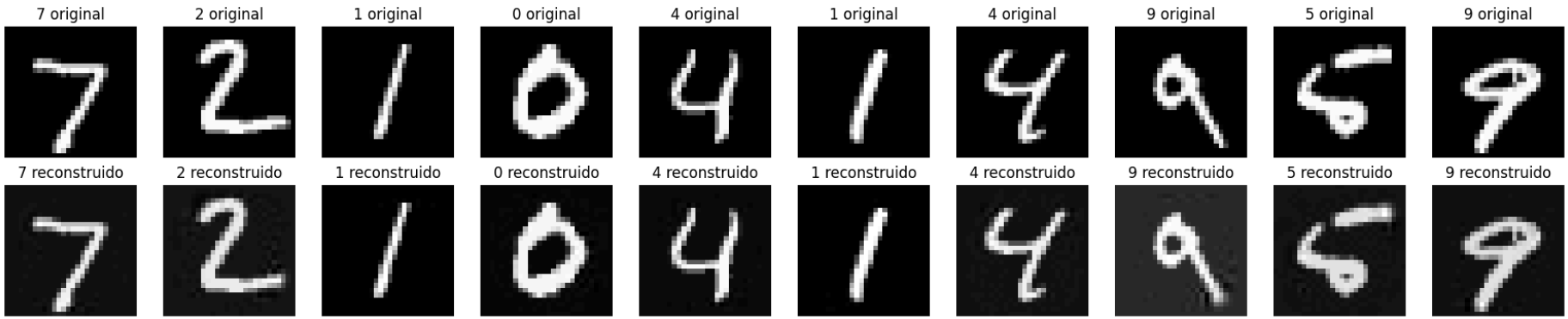


Eigendigits y reconstrucción: con $K = 400$

In [ ]:
```
K = 400; pca = PCA(n_components=K).fit(X_train); nrows, ncols = 1, 10
fig, axs = plt.subplots(nrows=nrows, ncols=ncols, figsize=(18, 18*nrows/ncols), constrained_layout=True)
for i in range(min(K, nrows * ncols)):
    ax = axs.flat[i]; ax.set_axis_off(); ax.set_title(f"Eigendigit {i}")
    ax.imshow(pca.components_[i, :].reshape((28, 28)), cmap=plt.cm.gray, interpolation="none")
```

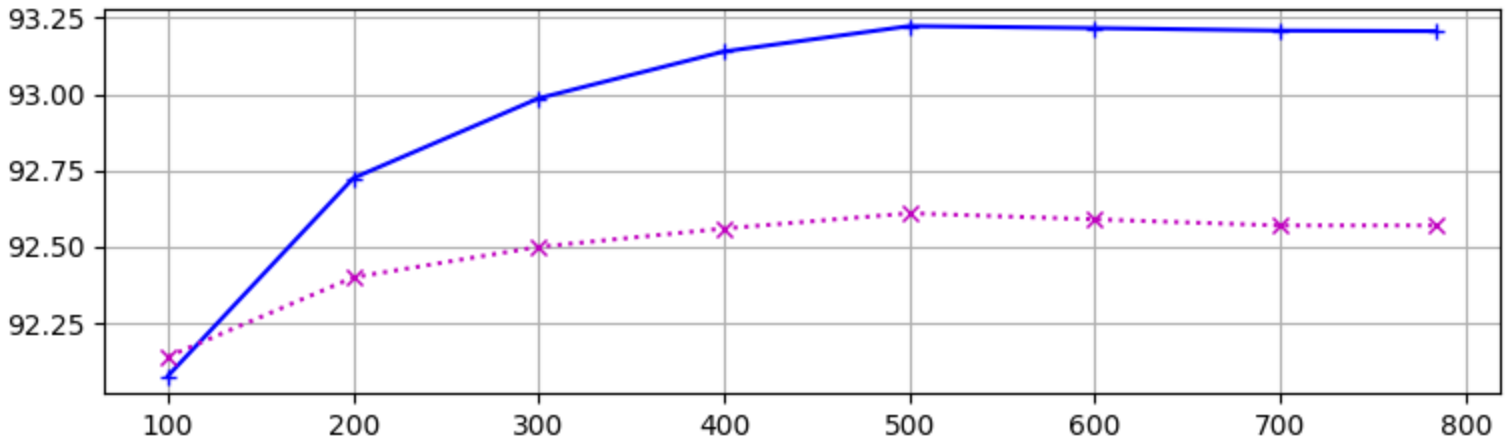


In [ ]:
```
ncols = 10; Z_test = pca.transform(X_test[:ncols]); hX_test = pca.inverse_transform(Z_test)
fig, axs = plt.subplots(nrows=2, ncols=10, figsize=(18, 18*2/ncols), constrained_layout=True)
for i in range(ncols):
    ax = axs.flat[i]; ax.set_axis_off(); ax.set_title(f"{y_test[i]} original")
    ax.imshow(X_test[i, :].reshape((28, 28)), cmap=plt.cm.gray, interpolation="none")
    ax = axs.flat[ncols + i]; ax.set_axis_off(); ax.set_title(f"{y_test[i]} reconstruido")
    ax.imshow(hX_test[i, :].reshape((28, 28)), cmap=plt.cm.gray, interpolation="none")
```



Precisión de regresión logística: estudio en función de K

In [ ]:
```
from sklearn.linear_model import LogisticRegression
max_K = np.min(X_train.shape); pca = PCA(n_components=max_K).fit(X_train)
Z_train = pca.transform(X_train); Z_test = pca.transform(X_test)
Ks = np.array([100, 200, 300, 400, 500, 600, 700, max_K])
acc_train = np.empty_like(Ks, dtype=float); acc_test = np.empty_like(Ks, dtype=float)
for i, K in enumerate(Ks):
    clf = LogisticRegression(C=0.1, max_iter=200).fit(Z_train[:, :K], y_train)
    acc_train[i] = clf.score(Z_train[:, :K], y_train)
    acc_test[i] = clf.score(Z_test[:, :K], y_test)
plt.figure(figsize=(9, 2.5)); plt.grid(True)
plt.plot(Ks, 100. * acc_train, '+-b', Ks, 100. * acc_test, 'x:m');
```



5 Fashion-MNIST

Fashion MNIST: MNIST con prendas de ropa; introducido en 2017 como sustituto del ya agotado MNIST

In [ ]:
```
import numpy as np; import datasets
ds = datasets.load_dataset("zalando-datasets/fashion_mnist").with_format("numpy"); ds
```

Out[ ]:
```
DatasetDict({
    train: Dataset({
        features: ['image', 'label'],
        num_rows: 60000
    })
    test: Dataset({
        features: ['image', 'label'],
        num_rows: 10000
    })
})
```

In [ ]:
```
import matplotlib.pyplot as plt; nrows = 4; ncols = 15; N = nrows * ncols
labels = ("T-shirt/top", "Trouser", "Pullover", "Dress", "Coat", "Sandal", "Shirt", "Sneaker", "Bag", "Ankle boot")
_, axs = plt.subplots(nrows=nrows, ncols=ncols, figsize=(16, 16*nrows/ncols), constrained_layout=True)
for ax, x, y in zip(axs.flat, ds['train'][:N]['image'], ds['train'][:N]['label']):
    ax.set_axis_off(); ax.imshow(x, cmap=plt.cm.gray_r, interpolation="none"); ax.set_title(labels[y])
```

6 CIFAR-10

CIFAR-10: dataset de $60\,000$ imágenes 32×32 a color de 10 clases; 6000 por clase

In [ ]:
```python
import numpy as np; import datasets
ds = datasets.load_dataset("uoft-cs/cifar10").with_format("numpy"); ds
```

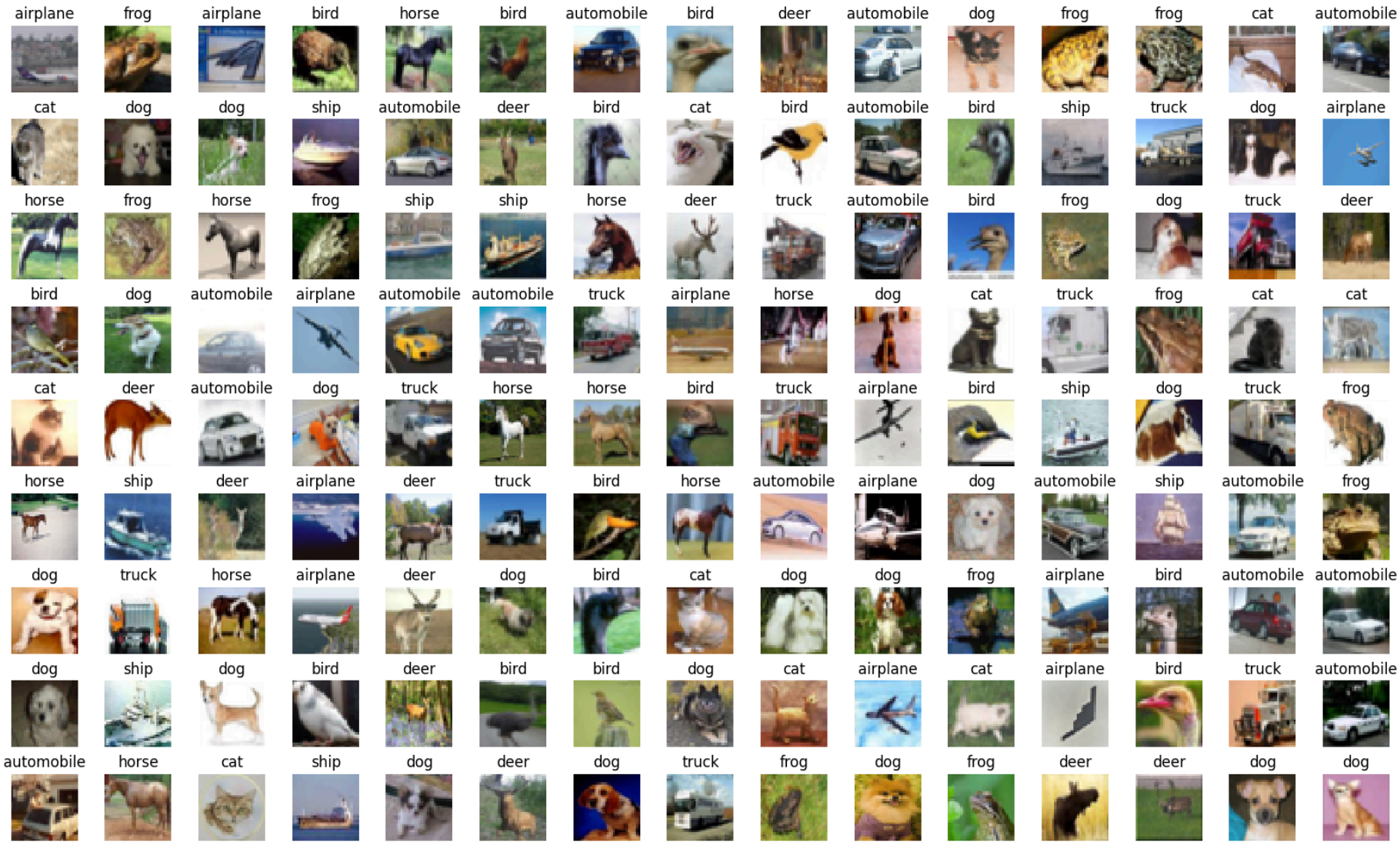
Out[ ]:
```
DatasetDict({
    train: Dataset({
        features: ['img', 'label'],
        num_rows: 50000
    })
    test: Dataset({
        features: ['img', 'label'],
        num_rows: 10000
    })
})
```

In [ ]:
```python
train0 = ds['train'][0]
train0['img'].shape, train0['img'].dtype, train0['label']
```

Out[ ]:
```
((32, 32, 3), dtype('uint8'), np.int64(0))
```

In [ ]:

```
import matplotlib.pyplot as plt; nrows = 9; ncols = 15; N = nrows * ncols
labels = ("airplane", "automobile", "bird", "cat", "deer", "dog", "frog", "horse", "ship", "truck")
_, axs = plt.subplots(nrows=nrows, ncols=ncols, figsize=(16, 16*nrows/ncols), constrained_layout=True)
for ax, x, y in zip(axs.flat, ds['train'][:N]['img'], ds['train'][:N]['label']):
    ax.set_axis_off(); ax.imshow(x, cmap=plt.cm.gray_r, interpolation="none"); ax.set_title(labels[y])
```



7 Ejercicio

Ejercicio: aplica PCA a Fashion-MNIST y CIFAR-10 siguiendo el ejemplo dado para MNIST

Aplicación a Fashion-MNIST: en comparación con el ejemplo dado para MNIST

- Comprueba que, para obtener un error reconstrucción prácticamente nulo, $K \geq 600$ (aprox.)
- Representa los diez primeros eigenarticles con $K = 600$
- Compara las diez primeras imágenes de test con sus reconstrucciones usando $K = 600$
- Comprueba que regresión logística supera ligeramente el 84% de precisión en test con $K \geq 200$

Aplicación a CIFAR-10: en comparación con el ejemplo dado para MNIST

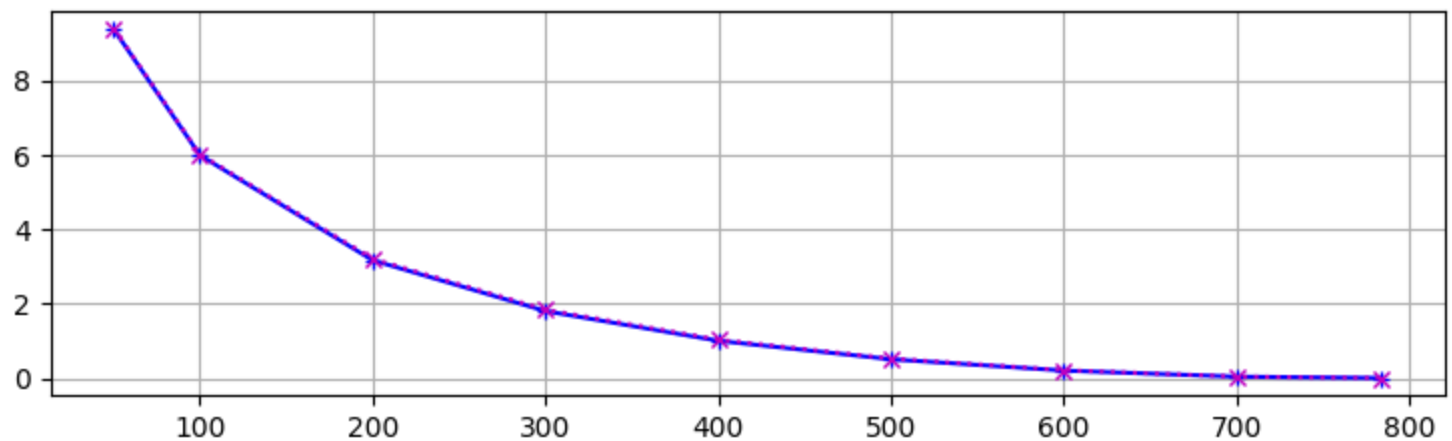
- Comprueba que, para obtener un error reconstrucción prácticamente nulo, $K \geq 1000$ (aprox.)
- No trates de representar gráficamente los diez primeros eigenobjects (requiere un PCA por cada canal de color)
- Compara las diez primeras imágenes de test con sus reconstrucciones usando $K = 1000$
- Comprueba que regresión logística obtiene, aproximadamente, un 41% de precisión en test con $K \geq 200$

Fashion-MNIST

Error de reconstrucción: estudio en función de K

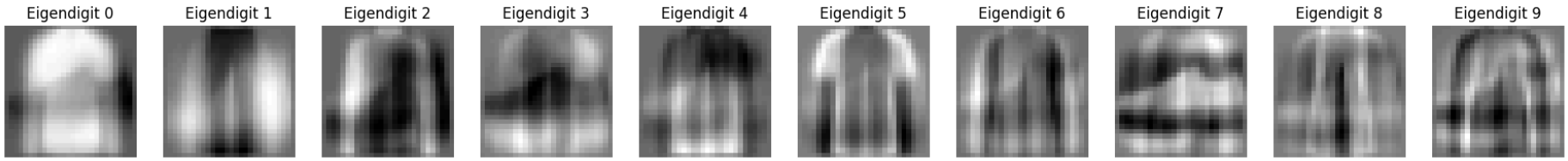
In [ ]:
```
import numpy as np; import datasets
ds = datasets.load_dataset("zalando-datasets/fashion_mnist").with_format("numpy")
X_train = ds['train'][:]['image'].astype(np.float32).reshape(-1, 28*28) / 255.
y_train = ds['train'][:]['label'].astype(np.uint8)
X_test = ds['test'][:]['image'].astype(np.float32).reshape(-1, 28*28) / 255.
y_test = ds['test'][:]['label'].astype(np.uint8)
labels = ("T-shirt/top", "Trouser", "Pullover", "Dress", "Coat", "Sandal", "Shirt", "Sneaker", "Bag", "Ankle boot")
```

In [ ]:
```
import matplotlib.pyplot as plt; from sklearn.decomposition import PCA
max_K = np.min(X_train.shape); pca = PCA(n_components=max_K).fit(X_train)
Z_train = pca.transform(X_train); Z_test = pca.transform(X_test)
Ks = np.array([50, 100, 200, 300, 400, 500, 600, 700, max_K])
L_train = np.empty_like(Ks, dtype=float); L_test = np.empty_like(Ks, dtype=float)
for i, K in enumerate(Ks):
    Z_train_K = Z_train.copy(); Z_train_K[:, K:] = 0.0; hX_train = pca.inverse_transform(Z_train_K)
    L_train[i] = np.square(X_train - hX_train).sum(axis=1).mean()
    Z_test_K = Z_test.copy(); Z_test_K[:, K:] = 0.0; hX_test = pca.inverse_transform(Z_test_K)
    L_test[i] = np.square(X_test - hX_test).sum(axis=1).mean()
plt.figure(figsize=(9, 2.5)); plt.grid(True); plt.plot(Ks, L_train, '+-b', Ks, L_test, 'x:m');
```

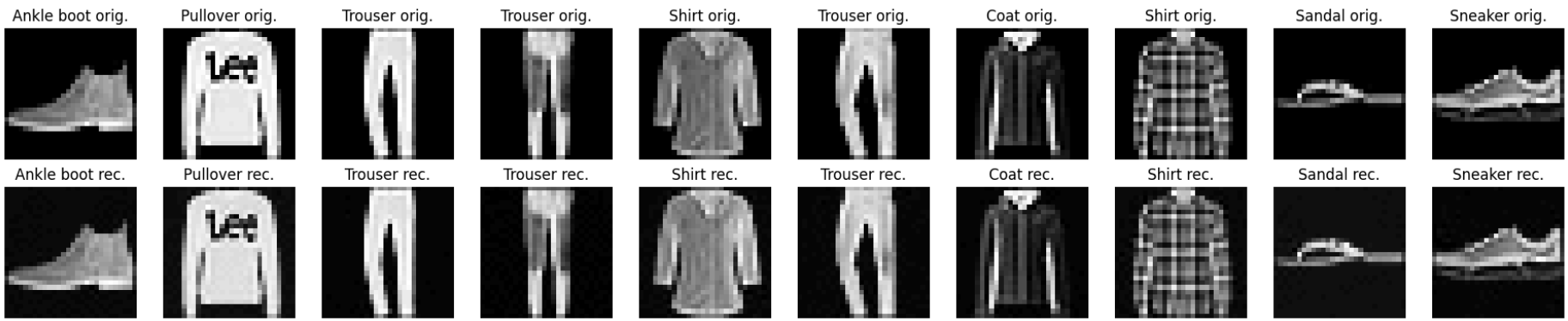


Eigenarticles y reconstrucción: con $K = 600$

In [ ]:
```
K = 600; pca = PCA(n_components=K).fit(X_train); nrows, ncols = 1, 10
fig, axs = plt.subplots(nrows=nrows, ncols=ncols, figsize=(18, 18*nrows/ncols), constrained_layout=True)
for i in range(min(K, nrows * ncols)):
    ax = axs.flat[i]; ax.set_axis_off(); ax.set_title(f"Eigendigit {i}")
    ax.imshow(pca.components_[i, :].reshape((28, 28)), cmap=plt.cm.gray, interpolation="none")
```

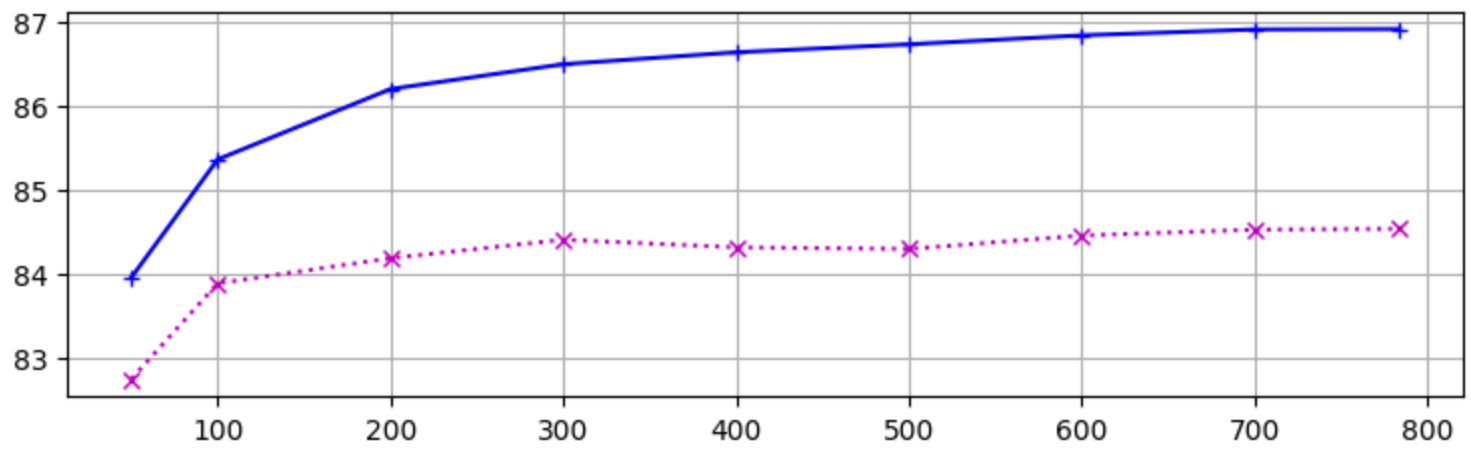


In [ ]:
```
ncols = 10; Z_test = pca.transform(X_test[:ncols]); hX_test = pca.inverse_transform(Z_test)
fig, axs = plt.subplots(nrows=2, ncols=10, figsize=(18, 18*2/ncols), constrained_layout=True)
for i in range(ncols):
    ax = axs.flat[i]; ax.set_axis_off(); ax.set_title(f"{labels[y_test[i]]} orig.")
    ax.imshow(X_test[i, :].reshape((28, 28)), cmap=plt.cm.gray, interpolation="none")
    ax = axs.flat[ncols + i]; ax.set_axis_off(); ax.set_title(f"{labels[y_test[i]]} rec.")
    ax.imshow(hX_test[i, :].reshape((28, 28)), cmap=plt.cm.gray, interpolation="none")
```



Precisión de regresión logística: estudio en función de K

In [ ]:
```
from sklearn.linear_model import LogisticRegression
max_K = np.min(X_train.shape); pca = PCA(n_components=max_K).fit(X_train)
Z_train = pca.transform(X_train); Z_test = pca.transform(X_test)
Ks = np.array([50, 100, 200, 300, 400, 500, 600, 700, max_K])
acc_train = np.empty_like(Ks, dtype=float); acc_test = np.empty_like(Ks, dtype=float)
for i, K in enumerate(Ks):
    clf = LogisticRegression(C=0.05, max_iter=200).fit(Z_train[:, :K], y_train)
    acc_train[i] = clf.score(Z_train[:, :K], y_train)
    acc_test[i] = clf.score(Z_test[:, :K], y_test)
plt.figure(figsize=(9, 2.5)); plt.grid(True)
plt.plot(Ks, 100. * acc_train, '+-b', Ks, 100. * acc_test, 'x:m');
```

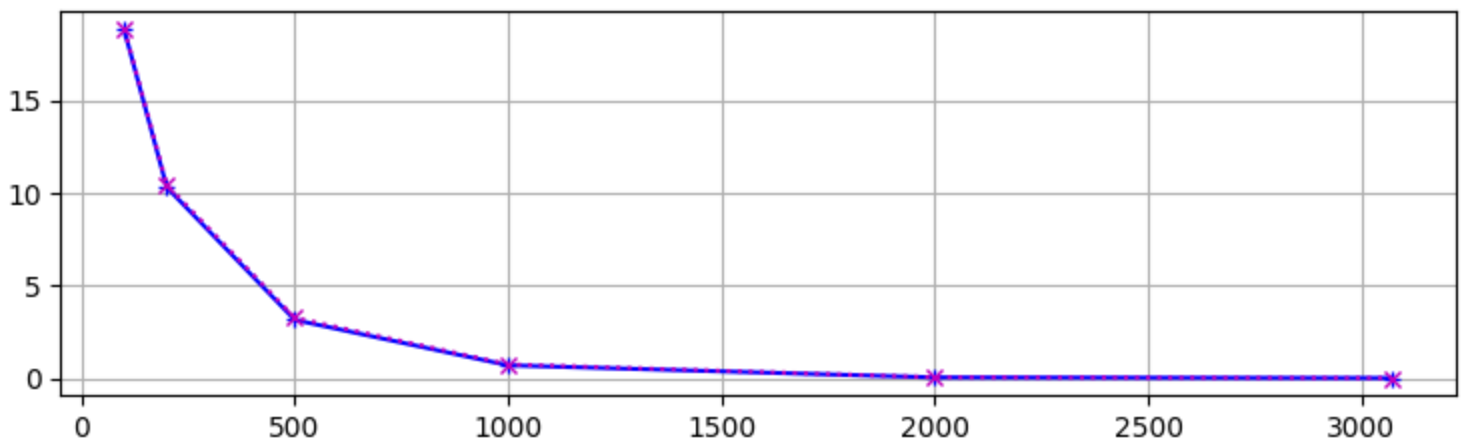


CIFAR-10

Error de reconstrucción: estudio en función de K

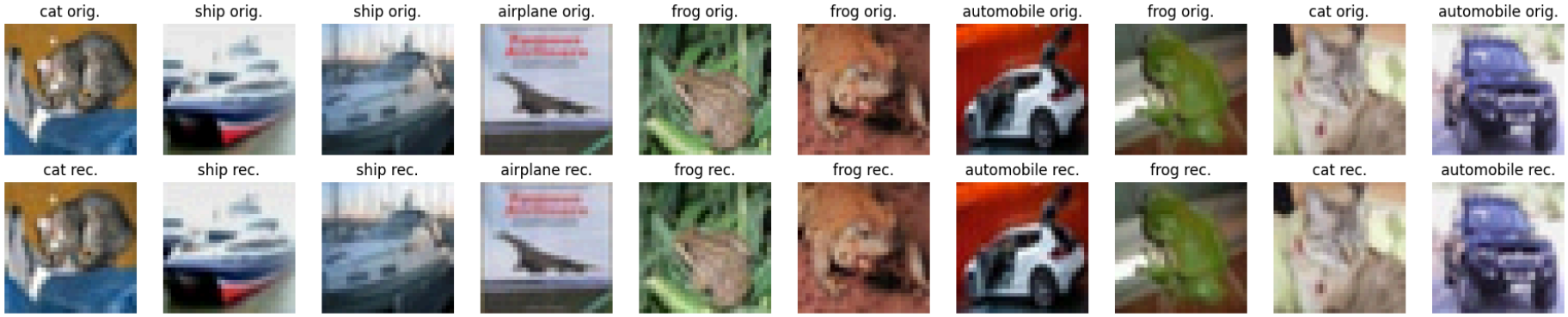
In [ ]:
```
import numpy as np; import datasets
ds = datasets.load_dataset("uoft-cs/cifar10").with_format("numpy")
X_train = ds['train'][:]['img'].astype(np.float32).reshape(-1, 32*32*3) / 255.
y_train = ds['train'][:]['label'].astype(np.uint8)
X_test = ds['test'][:]['img'].astype(np.float32).reshape(-1, 32*32*3) / 255.
y_test = ds['test'][:]['label'].astype(np.uint8)
labels = ("airplane", "automobile", "bird", "cat", "deer", "dog", "frog", "horse", "ship", "truck")
```

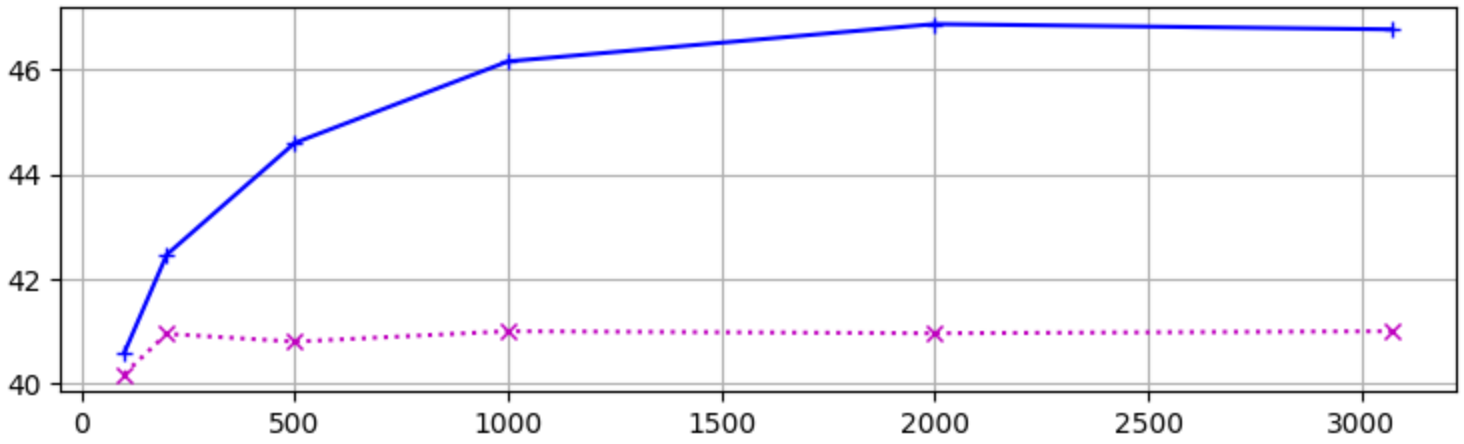
In [ ]:
```
import matplotlib.pyplot as plt; from sklearn.decomposition import PCA
max_K = np.min(X_train.shape); pca = PCA(n_components=max_K).fit(X_train)
Z_train = pca.transform(X_train); Z_test = pca.transform(X_test)
Ks = np.array([100, 200, 500, 1000, 2000, max_K])
L_train = np.empty_like(Ks, dtype=float); L_test = np.empty_like(Ks, dtype=float)
for i, K in enumerate(Ks):
    Z_train_K = Z_train.copy(); Z_train_K[:, K:] = 0.0; hX_train = pca.inverse_transform(Z_train_K)
    L_train[i] = np.square(X_train - hX_train).sum(axis=1).mean()
    Z_test_K = Z_test.copy(); Z_test_K[:, K:] = 0.0; hX_test = pca.inverse_transform(Z_test_K)
    L_test[i] = np.square(X_test - hX_test).sum(axis=1).mean()
plt.figure(figsize=(9, 2.5)); plt.grid(True); plt.plot(Ks, L_train, '+-b', Ks, L_test, 'x:m');
```



Reconstrucción: con $K = 1000$

In [ ]:
```
K = 1000; pca = PCA(n_components=K).fit(X_train)
ncols = 10; Z_test = pca.transform(X_test[:ncols]); hX_test = pca.inverse_transform(Z_test)
fig, axs = plt.subplots(nrows=2, ncols=ncols, figsize=(18, 18*2/ncols), constrained_layout=True)
for i in range(ncols):
    ax = axs.flat[i]; ax.set_axis_off(); ax.set_title(f"{labels[y_test[i]]} orig.")
    ax.imshow(X_test[i, :].reshape((32, 32, 3)), cmap=plt.cm.gray, interpolation="none")
    ax = axs.flat[ncols + i]; ax.set_axis_off(); ax.set_title(f"{labels[y_test[i]]} rec.")
    ax.imshow(hX_test[i, :].reshape((32, 32, 3)), cmap=plt.cm.gray, interpolation="none")
```



Precisión de regresión logística: estudio en función de K

In [ ]:
```
from sklearn.linear_model import LogisticRegression
max_K = np.min(X_train.shape); pca = PCA(n_components=max_K).fit(X_train)
Z_train = pca.transform(X_train); Z_test = pca.transform(X_test)
Ks = np.array([100, 200, 500, 1000, 2000, max_K])
acc_train = np.empty_like(Ks, dtype=float); acc_test = np.empty_like(Ks, dtype=float)
for i, K in enumerate(Ks):
    clf = LogisticRegression(C=0.05, max_iter=200).fit(Z_train[:, :K], y_train)
    acc_train[i] = clf.score(Z_train[:, :K], y_train)
    acc_test[i] = clf.score(Z_test[:, :K], y_test)
plt.figure(figsize=(9, 2.5)); plt.grid(True)
plt.plot(Ks, 100. * acc_train, '+-b', Ks, 100. * acc_test, 'x:m');
```